

Working with IntelliJ – Part 3

Student Workbook 6c – Integrated Development Environment

Version 7.1 Y

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Module 1

Working with JAR files in IntelliJ

Section 1–1

Working with JAR Files

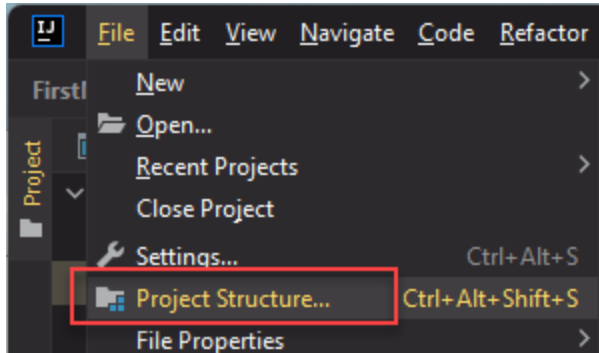
Exporting a Jar File in IntelliJ

- A Jar file is short for Java Archive. It is merely an enhanced zip or tar file that contains Java class files
- Jar files are portable, and can be transferred over the wire and even be included in other projects
 - They are portable largely because they are a single file, instead of a collection of dozens or more files
- A special type of jar file is a *Runnable Jar File*. What makes it runnable is a `MANIFEST.MF` that is included in the jar file. The `MANIFEST.MF` file contains an entry called `Main-Class` which will tell the Java Runtime Environment what class's main method to start.

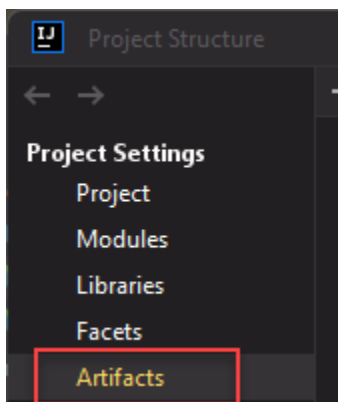
`Main-Class: com.mycompany.MainApp`

Creating a Jar File in IntelliJ

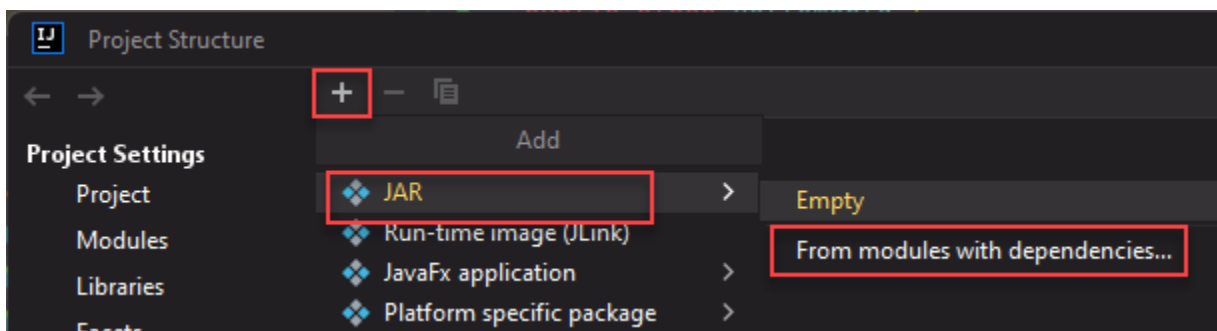
- Click File -> Project Structure



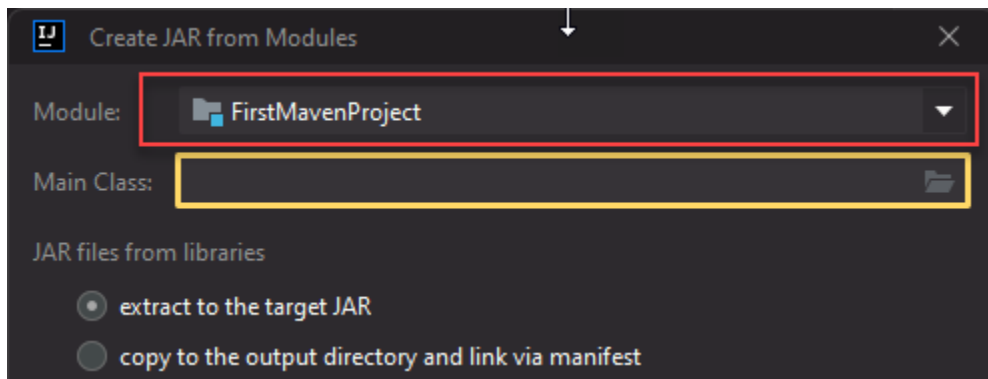
- Select Project Settings -> Artifacts



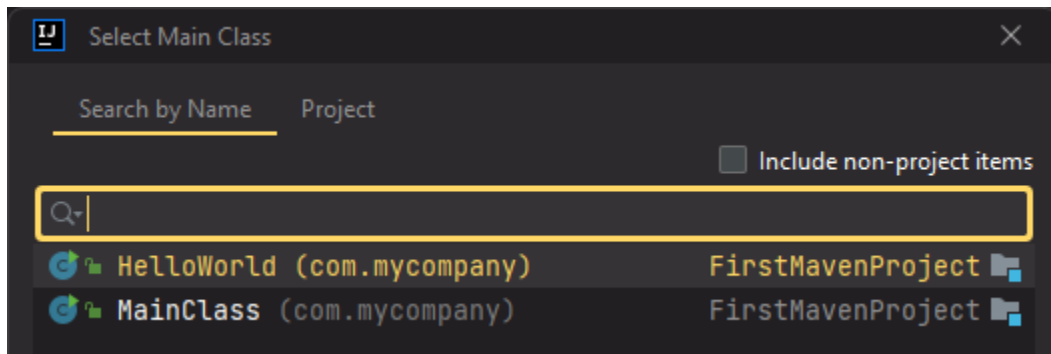
- Click the + to create a new artifact and select Jar -> From modules with dependencies



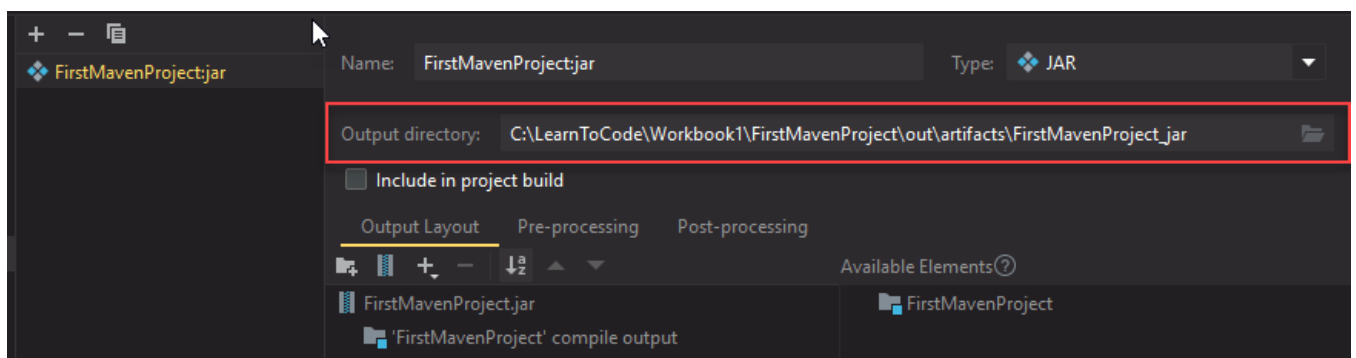
- Ensure that your Project is selected



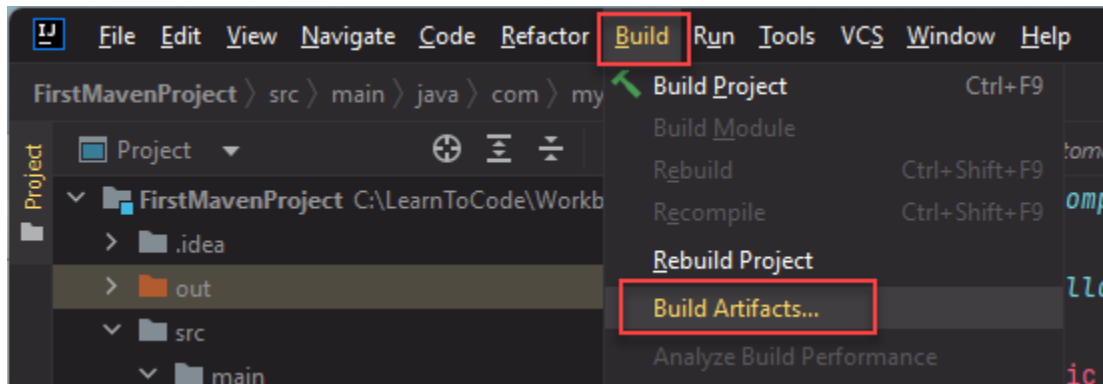
- Select the Main Class (this is the Entry Point for the Jar)



- Click OK
- This does not generate the Jar, but configures where the Jar will be created



- Click OK to save the configuration
- To generate the Jar go to Build -> Build Artifacts

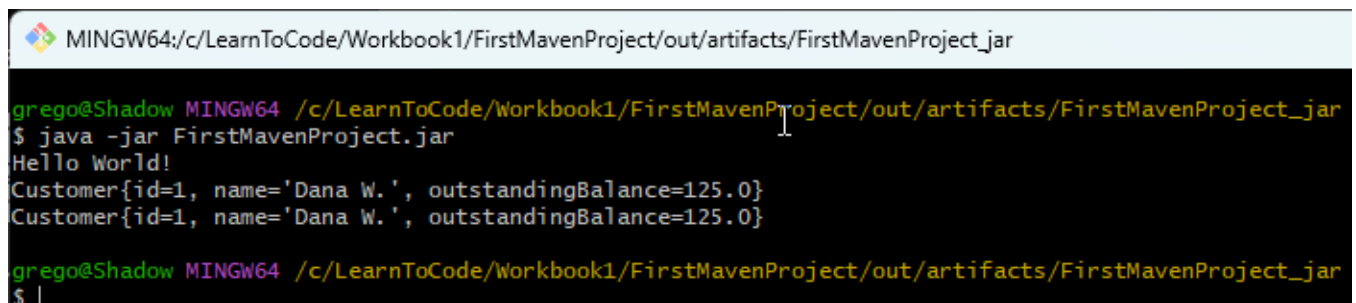


Running a Jar File on the Command Line

- Once you have the Jar file, open a Command Line and run the command

```
java -jar <jar-file>
```

and you should see the same results as you did in IntelliJ.



The screenshot shows a Windows Command Prompt window with the title bar "MINGW64:/c/LearnToCode/Workbook1/FirstMavenProject/out/artifacts/FirstMavenProject_jar". The prompt is "grego@Shadow MINGW64 /c/LearnToCode/Workbook1/FirstMavenProject/out/artifacts/FirstMavenProject_jar". The user enters the command "\$ java -jar FirstMavenProject.jar". The output is "Hello World!" followed by two lines of JSON: "Customer{id=1, name='Dana W.', outstandingBalance=125.0}" and "Customer{id=1, name='Dana W.', outstandingBalance=125.0}". The prompt then returns to "\$ |".

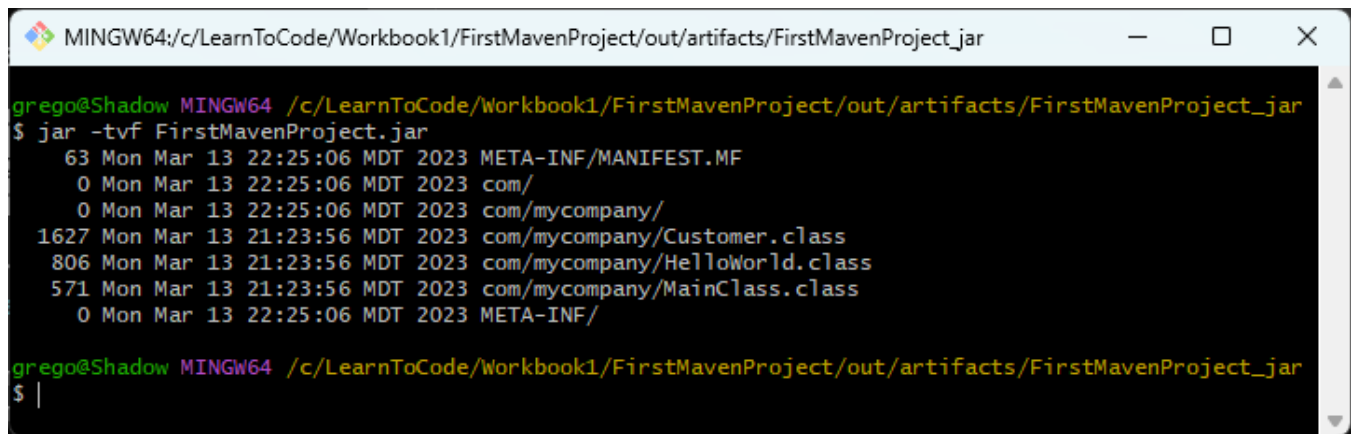
```
grego@Shadow MINGW64 /c/LearnToCode/Workbook1/FirstMavenProject/out/artifacts/FirstMavenProject_jar
$ java -jar FirstMavenProject.jar
Hello World!
Customer{id=1, name='Dana W.', outstandingBalance=125.0}
Customer{id=1, name='Dana W.', outstandingBalance=125.0}
grego@Shadow MINGW64 /c/LearnToCode/Workbook1/FirstMavenProject/out/artifacts/FirstMavenProject_jar
$ |
```

Viewing the Contents of the Jar File

- To view the contents of the jar file, run the command

```
jar -tvf <jar-file>
```

- Here you will see the manifest file that contains the Main-class declaration



The screenshot shows a terminal window titled "MINGW64:/c:/LearnToCode/Workbook1/FirstMavenProject/out/artifacts/FirstMavenProject.jar". The user has entered the command `jar -tvf FirstMavenProject.jar`. The output lists the contents of the jar file, including a manifest file and three class files.

```
grego@Shadow MINGW64 /c:/LearnToCode/Workbook1/FirstMavenProject/out/artifacts/FirstMavenProject.jar
$ jar -tvf FirstMavenProject.jar
 63 Mon Mar 13 22:25:06 MDT 2023 META-INF/MANIFEST.MF
  0 Mon Mar 13 22:25:06 MDT 2023 com/
  0 Mon Mar 13 22:25:06 MDT 2023 com/mycompany/
1627 Mon Mar 13 21:23:56 MDT 2023 com/mycompany/Customer.class
 806 Mon Mar 13 21:23:56 MDT 2023 com/mycompany/HelloWorld.class
 571 Mon Mar 13 21:23:56 MDT 2023 com/mycompany/MainClass.class
  0 Mon Mar 13 22:25:06 MDT 2023 META-INF/

grego@Shadow MINGW64 /c:/LearnToCode/Workbook1/FirstMavenProject/out/artifacts/FirstMavenProject.jar
$ |
```

- **NOTE:** You can also extract the .jar file with any Zip software and explore the contents directly

Exercises – Turtle Logo

EXERCISE 1

Open the turtle-logo project that you created in the `workbook-6` directory. Create an executable JAR file. To do this follow the workbook instructions to create a MANIFEST.MF in IntelliJ. This is the file that specifies your application's entry point.

Build and generate the JAR file.

Copy the JAR file to a new folder and run the application. You should be able to copy this to any folder on your computer and run the program from the JAR.